

Milestone-1 Submission Report

Project Title: AI Powered Smart Email Classifier for Enterprises

Milestone: Milestone–1 (Data Collection & Preprocessing)

1. Introduction

The purpose of this project is to build an AI-powered system that automatically classifies customer emails into predefined categories and assigns urgency levels. Milestone-1 focuses on collecting the dataset, cleaning the data, and preparing it for model training.

2. Dataset Collection

- The dataset was taken from the Consumer Complaint Database available on Kaggle.
- The dataset contains consumer complaints related to finance, credit reporting, loans, banking.
- Raw file used: rows.csv (863 MB)

3. Data Loading

The dataset was uploaded into Google Colab for preprocessing.

4. Data Cleaning & Column Selection

Selected columns:

- Product → category
- Issue → subcategory
- Consumer complaint narrative → text

Missing complaint rows were removed.

Final usable rows: 482,826

5. Column Renaming

Columns renamed:

- text

- category
- subcategory

6. Text Preprocessing

Performed:

- Lowercasing
- Removing HTML tags
- Removing URLs
- Removing numbers & punctuation
- Removing extra spaces
- Stopwords removal

7. Final Output File

cleaned_complaints.csv was generated and saved.

8. Milestone-1 Completion Summary

- Dataset collected
- Data loaded
- Columns selected
- Missing values removed
- Text cleaned
- Stopwords removed
- Final dataset exported

Milestone-1 is successfully completed.